

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv("https://raw.githubusercontent.com/shrikant-temburwar/Iris-Dataset/
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Id               150 non-null   int64
1   SepalLengthCm    150 non-null   float64
2   SepalWidthCm     150 non-null   float64
3   PetalLengthCm    150 non-null   float64
4   PetalWidthCm     150 non-null   float64
5   Species          150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
In [5]: df.describe()
```

```
Out[5]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
In [6]: df.shape
```

```
Out[6]: (150, 6)
```

```
In [7]: df["Species"].unique()
```

```
Out[7]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

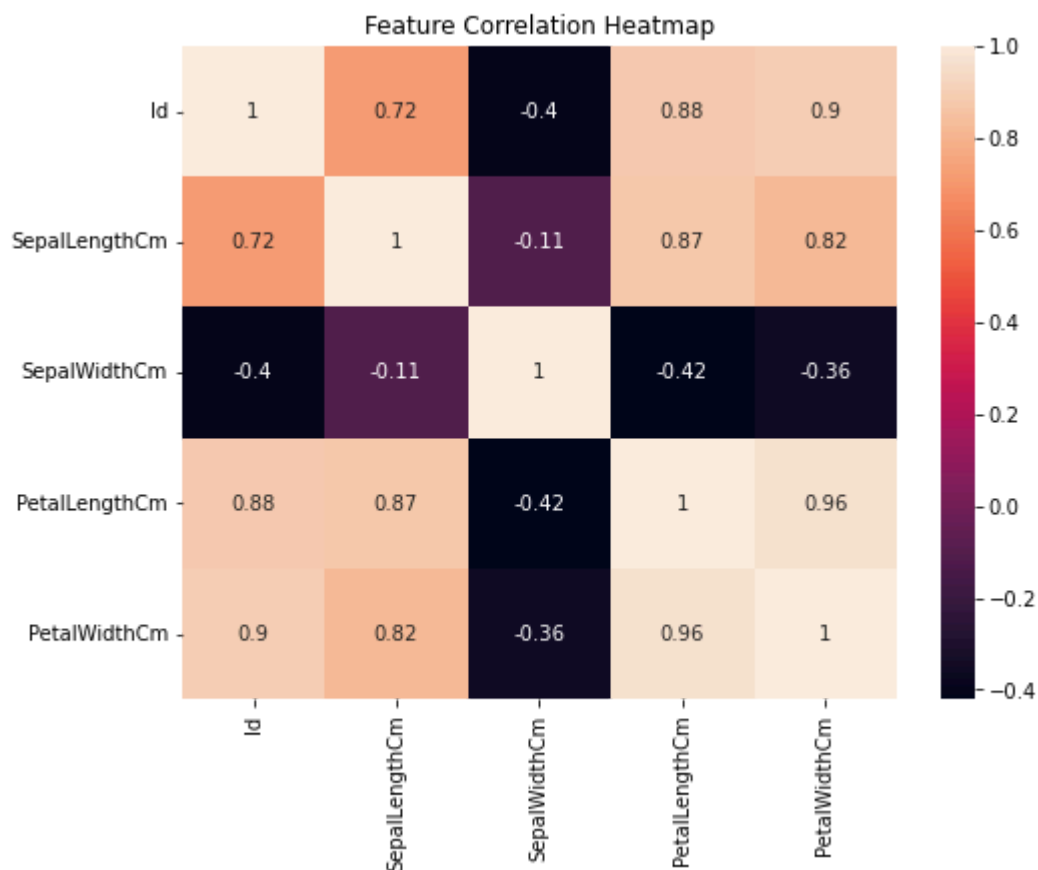
```
In [8]: df.groupby("Species").size()
```

```
Out[8]: Species
Iris-setosa      50
Iris-versicolor  50
Iris-virginica   50
dtype: int64
```

```
In [9]: df.dtypes
```

```
Out[9]: Id                int64
SepalLengthCm    float64
SepalWidthCm      float64
PetalLengthCm     float64
PetalWidthCm      float64
Species          object
dtype: object
```

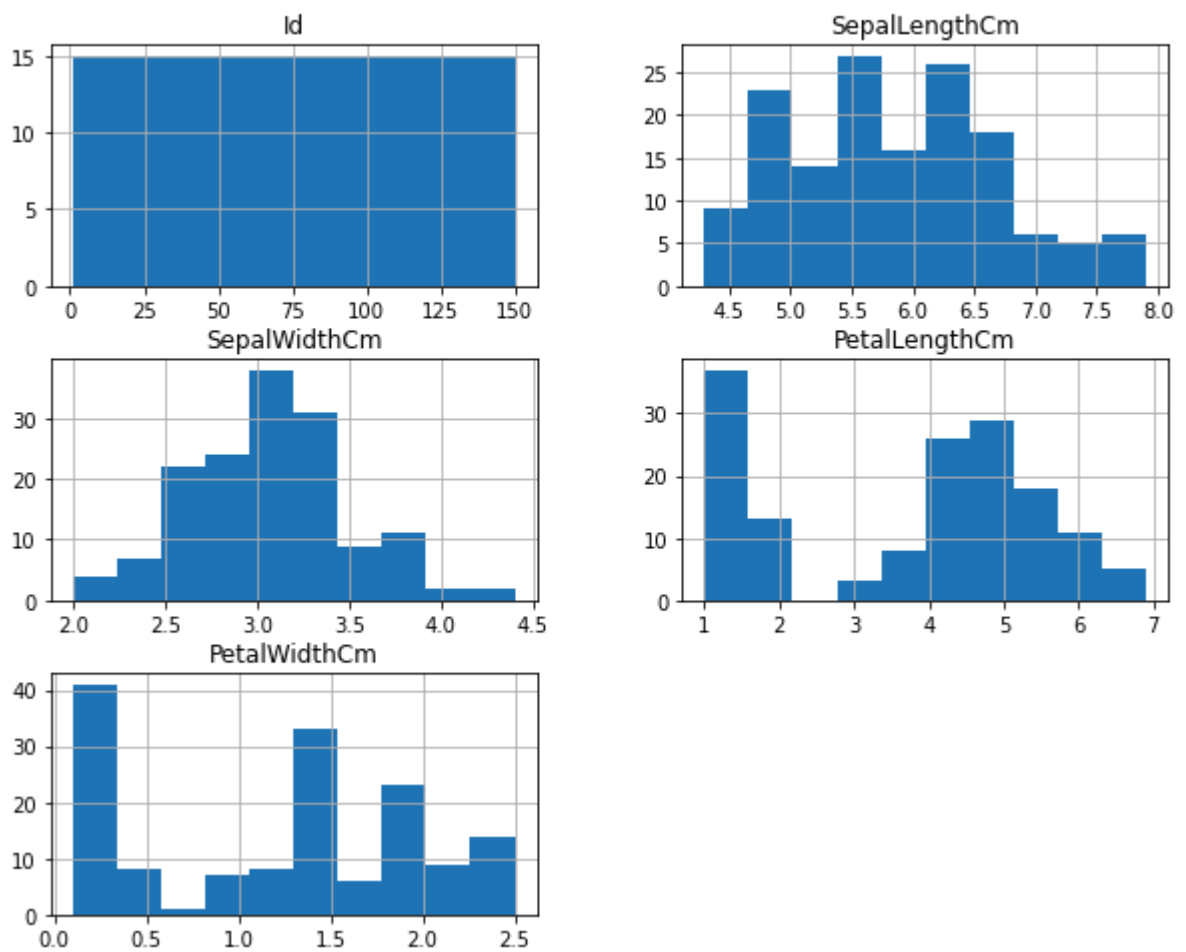
```
In [10]: plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Feature Correlation Heatmap")
plt.show()
```



```
In [11]: df.hist(figsize=(10,8))
plt.suptitle("Feature Distribution")
```

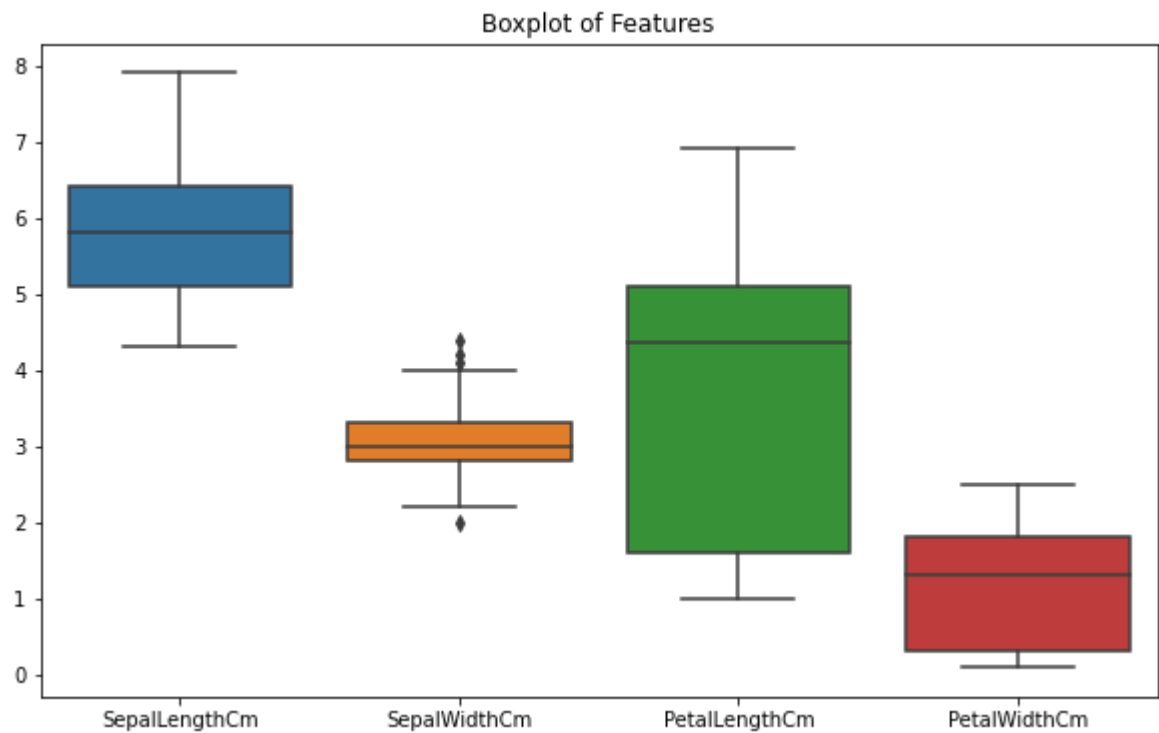
```
plt.show()
```

Feature Distribution



```
In [15]: plt.figure(figsize=(10,6))
sns.boxplot(data=df.drop(["Id", "Species"], axis=1))
plt.title("Boxplot of Features")
plt.show()
```

C:\Users\kolig\anaconda3\lib\site-packages\seaborn\categorical.py:82: FutureWarning: iteritems is deprecated and will be removed in a future version. Use .items instead.
 plot_data = [np.asarray(s, float) for k, s in iter_data]



```
In [13]: import numpy as np

for col in df.select_dtypes(include=np.number).columns:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower) | (df[col] > upper)]

    print(f"{col}: {len(outliers)} outliers")
```

```
Id: 0 outliers
SepalLengthCm: 0 outliers
SepalWidthCm: 4 outliers
PetalLengthCm: 0 outliers
PetalWidthCm: 0 outliers
```

```
In [ ]:
```